MUHAMMAD ZHARFAN HAKIM, S.KOM.

+62 822-1498-3599 | jarrworkspace@gmail.com | linkedin.com/in/mzharfanhakim | Portfolio

Full-Stack Engineer with 2+ years of professional experience building AI-powered enterprise applications, engineering blueprint review systems, workflow automation, and Web3/e-commerce products. Strong in TypeScript, React, Next.js, TanStack, Node.js, PostgreSQL, and n8n, with hands-on experience integrating OpenAI and Google Gemini models into auditable, human-reviewable workflows. Experienced collaborating with teams in Japan, Thailand, and Indonesia.

PROFESSIONAL EXPERIENCE

80&Company - **Kyoto, Japan** Aug 2024 - Present

Full-Stack Engineer

AI Engineering Drawing Review Platform

- Developed an enterprise platform that combines OCR, multimodal AI, rule-based evaluation, and evidence-linked verdicts to review complex engineering blueprints.
- Built and versioned n8n workflows for multi-page blueprint processing, per-patch analysis, OCR reuse, batch execution, concurrent extraction, and structured result generation using OpenAI and Google Gemini.
- Implemented cross-validation, grounded explanations, geometric-tolerance reasoning, JIS-backed checks, and explicit inconclusive-result handling to improve review reliability.
- Added reusable observations, model-specific fallbacks, recovery gates, debug logging, and workflow-version tracking to make AI results easier to diagnose and reproduce.
- Created React interfaces for blueprint and patch overlays, evidence inspection, requirement-level comparisons, review progress, workflow history, CSV exports, and debug downloads.
- Built the frontend foundation with React 19, TypeScript, TanStack Router, TanStack Query, Tailwind CSS, Zod, Vitest, and generated API clients.
- Delivered authentication, password change, project creation and hierarchy, revision analysis, requirement selection, evidence comparison, notifications, and administrator screens.
- Designed PostgreSQL schemas and migrations for users, sessions, projects, blueprints, revision lineage, review jobs, audit events, token usage, workflow versions, and access controls.
- Strengthened session isolation, project authorization, protected actions, result validation, error handling, and automated integration coverage through iterative code review.

AI Audio & Transcription Platform - Research & Development

- Developed an AI-powered Next.js platform using OpenAI generative, Realtime, and speech-to-text models together with Amazon Transcribe.
- Built end-to-end real-time and batch audio pipelines for transcription, processing, and AI-generated responses.
- Integrated multiple audio and language-model services into a unified user workflow.

Additional Client Projects

- Implemented print-accurate Japanese pharmacy and government forms for cities and prefectures across Japan using reusable, data-driven layouts.
- Developed ATS migration tooling with encoding-aware CSV ingestion, deterministic fixtures, a canonical mapping DSL, three-way duplicate classification, and sandboxed worker execution.
- Maintained Web3 backends and delivered NFT ownership, royalty distribution, and TON-based reward features while collaborating with international teams of 4-6 engineers.

PT Inti Konten Indonesia - **Bandung, Indonesia** Jul 2024 - Aug 2024

Full-Stack Engineer

- Developed a computer-vision application using OpenCV to count people visible in CCTV footage.
- Built a Next.js dashboard and improved page load performance by approximately 40%.
- Collaborated with engineers to define application content and deliver product requirements.

SELECTED EARLIER PROJECTS

- **Sakeworld NFT (2024):** Developed shared trademark-ownership NFTs, royalty-payment batch distribution, and performance improvements for an e-commerce platform.
- **Meme Fantasy (2025):** Built scheduled batch reward distribution for game winners using the TON SDK, Next.js, NestJS, PostgreSQL, and cron jobs.

EDUCATION

Telkom University - **Bandung, Indonesia** Aug 2021 - Nov 2025
Bachelor of Computer Science - GPA: 3.42/4.00

- Helped organize a Japanese cultural event attended by more than 10,000 visitors in one day.

Red Hat Academy Feb 2024 - Sep 2024
Red Hat System Administration II (RH134)

LEADERSHIP & ACTIVITIES

Nihon no Matsuri 14 - **Bandung, Indonesia** Sep 2022 - Jul 2023
Decoration Division Staff

- Designed event layouts, decorations, posters, certificates, and attraction rules as part of a 17-member team.
- Coordinated five performers and managed the experience for more than 80 attraction visitors.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, SQL, HTML, CSS
Frontend: React, Next.js, TanStack Router, TanStack Query, Tailwind CSS, Shadcn UI, Radix UI, Framer Motion
Backend & Data: Node.js, NestJS, REST APIs, PostgreSQL, MySQL, Prisma, Supabase, Kysely, Zod
AI & Automation: OpenAI APIs, Google Gemini, Amazon Transcribe, n8n, OCR, multimodal analysis, prompt/workflow versioning
Cloud & Tools: AWS, Vercel, Git, GitHub, Linux, Bash, Docker, Vitest, Playwright, Postman, Figma

CERTIFICATIONS & LANGUAGES

Certifications: Red Hat System Administration II (RH134); Node.js Intermediate; React Basic; JavaScript Intermediate; Frontend Developer (React); REST API Intermediate
English: TOEIC 845 **Japanese:** JLPT N4